

Combinatorics PhD Exam, August 1995

1. Every simple planar graph has a vertex of degree at most 5.
2. This problem has two parts.
 - (a) State the max-flow min-cut theorem for networks.
 - (b) Show that the max-flow min-cut theorem implies the following version of Menger's Theorem.
The maximum number of edge disjoint directed paths between two given points s, t of a directed graph equals the minimum number of edges whose removal separates s from t .
3. We have an unlimited supply of bricks of length 1, each painted red, white or blue, and bricks of length 2, each painted green or yellow. Explicitly find a_n = the number of ways to choose a sequence of bricks of total length n .
4. State and prove Dilworth's Theorem for a finite poset.
5. Each of n gentlemen checks both his hat and his umbrella at a restaurant. Both the hats and the umbrellas are returned randomly. What is the probability that no man gets back both his hat and his umbrella?
6. **Warning: Suspected error.** The summation upper limit is clearly printed as m , but m is otherwise undefined and no conditions are given under which the displayed identity holds. The source reading has been preserved because the intended correction is not uniquely determined.

Prove that

$$\sum_{i=0}^m \binom{s}{i} \binom{t}{k-i} = \binom{s+t}{k}.$$

7. 6 identical black balls and $n - 6$ identical white balls are arranged in a row. How many ways are there to do this so that no 3 consecutive balls are black?
8. Define a q -Hamming code C as one whose parity-check matrix H has as its columns one non-zero vector from each 1-dimensional subspace of the vector space $\text{GF}(q)^r$.
 - (a) What is the length of C ?
 - (b) What is the minimum distance in C ?
 - (c) Prove that C is perfect.
9. This problem has two parts.
 - (a) State the Bruck–Ryser–Chowla theorem.
 - (b) Use this theorem to prove that a projective plane of order 6 cannot exist.